



BANNARI AMMAN INSTITUTE OF TECHNOLOGY

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22CH203 – UNIT 1- ELECTROCHEMISTRY

LECTURE MATERIAL – TOPIC - 6

Topic: Understanding the significance of electrochemical series

Electrochemical Series

Electrochemical series, also referred to as activity series, is a list that describes the arrangement of elements in the order of their increasing electrode potential values. The series has been established by measuring the potential of various electrodes versus standard hydrogen electrodes (SHE).

In electrochemical series, the electrodes (metals and non-metals) in contact with their ions are arranged on the basis of the values of their standard reduction or oxidation potentials. Standard electrode potential is obtained by measuring the voltage when the half-cell is connected to the standard hydrogen electrode under standard conditions.

Electropositive and Electronegative Elements

Elements (other than hydrogen) that display a greater tendency to lose electrons to their solution are usually categorized as electropositive. Similarly, elements that acquire electrons are said to be electronegative. They are usually below the element hydrogen in the series. In any case, if we look at the electrochemical series, we can figure out the order in which metals replace one another with their salts. So, electropositive metals generally replace hydrogen from acids.

Electrochemical Series

The electrochemical series is a simple way of visualising similar vs dissimilar metals. For roofers, you need to know what compatibility issues exist in regard to the material. In this chart, common roofing metals, as well as uncommon ones, are shown. In most basic terms, metals that exist further apart from each other on this scale will react with a higher propensity for corrosion than ones that are close together. (i.e., Zinc and Copper = far apart on the scale. This means that you would never have a copper pipe draining water onto a zinc-coated roof)

Half Reaction	Standard Potential (V)
↑ stronger oxidizing agent	
$\text{F}_2 + 2\text{e}^- \rightleftharpoons 2\text{F}^-$	+2.87
$\text{Pb}^{4+} + 2\text{e}^- \rightleftharpoons \text{Pb}^{2+}$	+1.67
$\text{Cl}_2 + 2\text{e}^- \rightleftharpoons 2\text{Cl}^-$	+1.36
$\text{O}_2 + 4\text{H}^+ + 4\text{e}^- \rightleftharpoons 2\text{H}_2\text{O}$	+1.23
$\text{Ag}^+ + \text{e}^- \rightleftharpoons \text{Ag}$	+0.80
$\text{Fe}^{3+} + \text{e}^- \rightleftharpoons \text{Fe}^{2+}$	+0.77
$\text{Cu}^{2+} + 2\text{e}^- \rightleftharpoons \text{Cu}$	+0.34
$2\text{H}^+ + 2\text{e}^- \rightleftharpoons \text{H}_2$	0.00
$\text{Pb}^{2+} + 2\text{e}^- \rightleftharpoons \text{Pb}$	-0.13
$\text{Fe}^{2+} + 2\text{e}^- \rightleftharpoons \text{Fe}$	-0.44
$\text{Zn}^{2+} + 2\text{e}^- \rightleftharpoons \text{Zn}$	-0.76
$\text{Al}^{3+} + 3\text{e}^- \rightleftharpoons \text{Al}$	-1.66
$\text{Mg}^{2+} + 2\text{e}^- \rightleftharpoons \text{Mg}$	-2.36
$\text{Li}^+ + \text{e}^- \rightleftharpoons \text{Li}$	-3.05
	↓ stronger reducing agent

Application of Electrochemical Series

1. Oxidizing and Reducing Strengths

The electrochemical series helps us to identify a good oxidizing agent or reducing agent. All the substances appearing on the top of the electrochemical series are good oxidizing agents, i.e., they have a positive value of standard reduction potential, whereas those appearing on the bottom of the electrochemical series are good reducing agents, i.e., they have a negative value of standard reduction potential. For example, the F_2 electrode with the standard reduction potential value of +2.87 is a strong oxidizing agent, and Li^+ with a standard reduction potential value of -3.05 volts is a strong reducing agent.

2. Calculation of Standard emf (E^0) of Electrochemical Cell

The standard emf of the cell is the sum of the standard reduction potential of the two half cells: reduction half cell and oxidation half cell

$$E^0_{\text{cell}} = E^0_{\text{red}} + E^0_{\text{ox}}$$

By convention, the standard oxidation potential is always expressed in terms of reduction potential.

Thus, standard oxidation potential (E^0_{ox}) = – standard reduction potential E^0_{red}

Therefore,

$E^\circ_{\text{cell}} = (\text{standard reduction potential of reduction half cell}) - (\text{standard reduction potential of oxidation half cell})$

As oxidation takes place at the anode, and reduction takes place at the cathode. Hence,

$$E^\circ_{\text{cell}} = E^\circ_{\text{cathode}} - E^\circ_{\text{anode}}$$

Example:

For a reaction, $2\text{Ag}^+ (\text{aq}) + \text{Cd} \rightarrow 2\text{Ag} + \text{Cd}^{+2}(\text{aq})$

The standard reduction potential given are: $\text{Ag}^+/\text{Ag} = 0.80$ volt, $\text{Cd}^{+2}/\text{Cd} = -0.40$ volt

From the reaction, we can see that Cd losses electrons and Ag^+ gains. Hence, oxidation half cell or anode is Cd.

Using the formula,

$$E^\circ_{\text{cell}} = E^\circ_{\text{cathode}} - E^\circ_{\text{anode}}$$

$$= 0.80 - (-0.40)$$

$$= 1.20 \text{ volt}$$

3. Predicting the Feasibility of Redox Reaction

Any redox reaction would occur spontaneously if the free energy change (ΔG) is negative. The free energy is related to cell emf in the following manner:

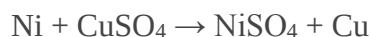
$$\Delta G^\circ = nFE^\circ$$

Where n is the number of electrons involved, F is the Faraday constant, and E° is the cell emf.

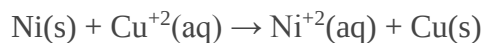
- ΔG° can be negative if E° is positive.
- When E° is positive, the cell reaction is spontaneous and serves as a source of electrical energy.
- If it comes out to be negative, then the spontaneous reaction cannot take place.
- The resultant value of E° for redox reaction is important in predicting the stability of a metal salt solution when stored in another metal container.

For example, let us find out whether we can store copper sulphate solution in a nickel vessel or not.

Given: $\text{Ni}^{+2}/\text{Ni} = -0.25$ volt, $\text{Cu}^{+2}/\text{Cu} = 0.34$ volt



We want to see whether Ni metal will displace copper from copper sulphate solution to give NiSO_4 by undergoing an oxidation reaction.



From the above reaction, it is clear the oxidation terminal will be the Ni electrode.

$$E^\circ_{\text{cell}} = E^\circ_{\text{cathode}} - E^\circ_{\text{anode}}$$

$$= 0.34 - (-0.25)$$

$$= 0.59 \text{ volt}$$

As the emf comes out to be positive, it implies that copper sulphate reacts when placed in a nickel vessel and hence cannot be stored in it.

4. Predicting the Product of Electrolysis

In case two or more types of positive and negative ions are present in the solution, during electrolysis, certain metal ions are discharged or liberated at the electrodes in preference to others. In general, in such competition, the ion, which is a stronger oxidizing agent (high value of standard reduction potential), is discharged first at the cathode.

Thus, when an aqueous solution of NaCl containing Na^+ , Cl^- , H^+ and OH^- ions is electrolysed, the H^+ ion is preferentially deposited at the cathode (reduction) instead of Na^+ being reduced; this is because the reduction potential of hydrogen (0.00 volt) is higher than the reduction potential of sodium (-2.71 volt). At the anode where oxidation takes place, the anion that has a lower reduction potential will be oxidized. Therefore, OH^- with a standard reduction potential of 0.40 volt will be oxidized in preference to Cl^- with a standard reduction potential of 1.36 volt.

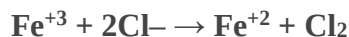
Electrochemical Series Important Points

- In the electrochemical series, the reduction potential of an element is taken in reference to the hydrogen scale where $E^\circ = \text{zero}$. As per the definition, the standard reduction potential of an element is described as the measure of the tendency of an element to undergo reduction.

- The greater the reduction potential of an element, the more easily it will be reduced. Meanwhile, elements that have low reduction potential will get oxidized more quickly and easily.
- Alternatively, elements that give up electrons without any difficulty have negative or lower reduction potential. Elements that do not give up electrons easily accept electrons effortlessly and have positive or higher reduction potential.
- Stronger reducing agents that have negative standard reduction potential are usually situated below the hydrogen in the electrochemical series. On the other hand, weaker reducing agents with positive standard reduction potential are found above the hydrogen in the series.
- As we move down in the group, the reducing agent's strength increases while the oxidizing agents' strength decreases.
- Likewise, as we move from top to bottom in the series, the electro-positivity and activity of metals amplify or intensify. In the case of non-metals, it decreases.

Solved Problems

1. Predict whether the following reaction will occur spontaneously or not:



$$\text{Fe}^{+2}/\text{Fe} = -0.440 \text{ volt} ; E^0\text{Cl} / \text{Cl}^- = 1.36 \text{ volt}$$

Solution:

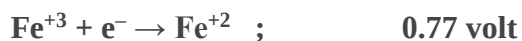
$$E^0_{\text{cell}} = E^0_{\text{cathode}} - E^0_{\text{anode}}$$

Since chlorine has a higher reduction potential than iron, at the cathode, reduction of chlorine occurs, and at the anode, oxidation of iron occurs.

$$E^0_{\text{cell}} = 1.36 - (-0.440) = 1.80 \text{ volts}$$

The positive value of E^0_{Cell} implies that the reaction occurs spontaneously.

2. The standard reduction potential at 25°C for the following half-reaction is given below:



Find the strongest reducing agent among the following

1. **Zn**
2. **Cr**
3. **Cu**
4. **Fe⁺³**

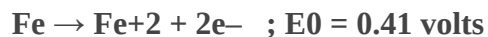
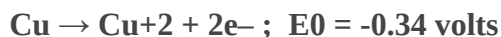
Solution: Option 1

A reducing agent is a chemical species that loses an electron to another chemical species in a redox chemical reaction. Since the reducing agent loses electrons, it is oxidized. Out of the following given half-reaction, the reduction of Zn⁺² has the lowest reduction potential (-0.762). We know that

Oxidation potential = -(reduction potential)

So, in terms of standard oxidation potential, zinc will have the highest oxidation potential, i.e., 0.762 volts. Therefore, zinc is the strongest reducing agent.

3. The standard oxidation potential, E₀, for the half-reactions are as follows:



Calculate the emf of the cell, $\text{Cu}^{+2} + \text{Fe} \rightarrow \text{Cu} + \text{Fe}^{+2}$

Solution:

$E^{\circ}_{\text{cell}} = (\text{standard reduction potential of reduction half cell}) - (\text{standard reduction potential of oxidation half cell})$

$E^{\circ}_{\text{cell}} = -(\text{standard oxidation potential of reduction half cell}) - (-\text{standard oxidation potential of oxidation half cell})$

$$E^{\circ}_{\text{cell}} = -0.34 - (-0.41)$$

$$E^{\circ}_{\text{cell}} = 0.07 \text{ volt.}$$

Reference:

[1] Jain and Jain, Engineering Chemistry, 16th Edition, Dhanpat Rai Publishing

Company, New Delhi, 2013

[2] P.H. Rieger, Electrochemistry, Second Edition (Reprint), Springer, Netherland,
2012

[3] S. Vairam, Engineering Chemistry, John Wiley & Sons, 2014

[4] Chemistry for Engineering students by Lawrence S Brown and Thomas A Holme
(Fourth Edition) Chapter 13